EE1-10B MATHEMATICS II

1. (a)

(i) Prove the frequency-shift property of the Fourier Transform $\mathscr{F}$, that if $\mathscr{F}[f(t)] = F(\omega)$, then [Bookwork]

$$\mathscr{F}\left[e^{iat} f(t)\right] = F(\omega - a)\,.$$

SOLUTION Using the definition of the Fourier Transform,

$$\mathscr{F}\left[e^{iat} f(t)\right] = \int_{-\infty}^{+\infty} e^{iat} f(t) e^{-i\omega t}\, dt = \int_{-\infty}^{+\infty} f(t) e^{-i(\omega - a)t}\, dt$$

and substituting $\bar{\omega} = \omega - a$ in the last integral, we get

$$\mathscr{F}\left[e^{iat} f(t)\right] = \int_{-\infty}^{+\infty} f(t) e^{-i\bar{\omega} t}\, dt = F(\bar{\omega}) = F(\omega - a)\,,$$

as required.

Errors were straight-forward, mostly from getting the variable substitution wrong.

(ii) Show that the Fourier Transform of the function [Seen similar. Hint: new]

$$g(t) = \begin{cases} |t| & -1 \le t \le 1 \\ 0 & \text{otherwise} \end{cases},$$

is $\mathscr{F}[g(t)] = 2\text{sinc}\ \omega - \text{sinc}^2\ (\omega/2)$, where $\text{sinc}\ x = \frac{\sin x}{x}$.

[Hint: consider $e^{-i\omega t} = \cos(\omega t) - i\sin(\omega t)$, and that $|t|$ is even. You may assume $\cos 2x = 1 - 2\sin^2 x$.]

SOLUTION It's not hard to obtain the FT with the usual exponential formula, but more work, than exploiting the even propery of g. Following the hint

$$\mathscr{F}[g(t)] = \int_{-\infty}^{+\infty} g(t)e^{-i\omega t}\,dt = \int_{-1}^{1} |t|\,(\cos\omega t - i\sin\omega t)\,dt = \int_{-1}^{1} |t|\,\cos\omega t\,dt - i\int_{-1}^{1} |t|\,\sin\omega t\,dt$$

In the last expression, the second integral is a symmetric integral of an odd function, so equal to zero. The first integral has an even integrand and we can rewrite as

$$\mathscr{F}[g(t)] = 2\int_0^1 t\,\cos\omega t\,dt = 2\left[t\frac{\sin\omega t}{\omega} + \frac{1}{\omega^2}\cos\omega t\right]_0^1 = 2\left[\frac{\sin\omega}{\omega} + \frac{1}{\omega^2}(\cos\omega - 1)\right]$$

$$= 2\mathrm{sinc}(\omega) - \frac{4}{\omega^2}\sin^2(\omega/2) = 2\,\mathrm{sinc}(\omega) - \mathrm{sinc}^2(\omega/2)$$

as required.

Errors were mostly in the integration by parts but they were minor ones. most of them were errors carried forward from initial mistakes.

(iii) Hence, or otherwise, compute $\mathscr{F}[g(t)\cos(at)]$. [bookwork/seen similar]

SOLUTION

This is modulation. We use $\cos at = \dfrac{e^{iat} + e^{-iat}}{2}$ and linearity. Then

$$\mathscr{F}[g(t)\cos(at)] = \mathscr{F}\left[g(t)\frac{1}{2}(e^{iat} + e^{-iat})\right] = \frac{1}{2}\mathscr{F}\left[g(t)e^{iat}\right] + \frac{1}{2}\mathscr{F}\left[g(t)e^{-iat}\right]$$

Now apply (i) and (ii):

$$\mathscr{F}[g(t)\cos(at)] = \frac{1}{2}G(\omega - a) + \frac{1}{2}G(\omega + a)$$

$$= \mathrm{sinc}(\omega - a) + \mathrm{sinc}(\omega + a) - \tfrac{1}{2}\mathrm{sinc}^2\left(\tfrac{\omega - a}{2}\right) - \tfrac{1}{2}\mathrm{sinc}^2\left(\tfrac{\omega + a}{2}\right).$$

Errors were trivial. Either students understood how to do or they didn't.

(b) Two planes have equations

$$\Pi_1 : 2x+y+z=1 \quad \text{and} \quad \Pi_2 : x-3y+z=2\,.$$

(i) Show that L, the line of intersection of Π_1 and Π_2, has equation

$$\underline{\mathbf{r}} = (-1,0,3)+\lambda(-4,1,7)\,. \qquad \text{[seen similar]}$$

SOLUTION

Eliminate x from $2x+y+z=1$ and $x-3y+z=2$ to get $7y-z=-3$. Two unknowns, one equation: there is a free variable. Let $y=\lambda\in\mathbb{R}$, then $z=3+7\lambda$ and $x=2-z+3y=-1-4\lambda$. In vector form, this is the equation of a line L in R^3:

$$L: \qquad \underline{\mathbf{r}} = \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ 1 \\ 7 \end{pmatrix}$$

Several students resorted to reverse-engineering the question to get to the solution, i.e. starting with what was given instead of showing the logical flow towards what had to be proven. for example, they assumed that the point $(-1,0,3)$ was chosen and then they showed "magically" that the right answer was obtained. This is unacceptable as the description of the steps did not follow the right logic.

(ii) Find Q, the point where the plane $\Pi_3 : x+y-z=6$ intersects the line L. [seen similar]

SOLUTION Substitute L into Π_3:

$$(-1-4\lambda)+(0+\lambda)-(3+7\lambda)=6 \Rightarrow \lambda=-1$$

giving the coordinates as

$$Q: \qquad \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} - \begin{pmatrix} -4 \\ 1 \\ 7 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ -4 \end{pmatrix}$$

Errors were mostly arithmetic in nature.

(iii) Obtain the distance between L and the point $P(1,3,-1)$. [seen similar]

SOLUTION

Let $P'(x,y,z)$ be the point on L nearest to $P(1,3,-1)$. Then $\overrightarrow{PP'}$ is perpendicular to L, so that

$$\overrightarrow{PP'} \cdot \begin{pmatrix} -4 \\ 1 \\ 7 \end{pmatrix} = \begin{pmatrix} x-1 \\ y-3 \\ z+1 \end{pmatrix} \cdot \begin{pmatrix} -4 \\ 1 \\ 7 \end{pmatrix} = 0 \Rightarrow -4x+y+7z = -8$$

The last equation is for the plane containing P and P' and perpendicular to L, so that P' is the intersection of L and this plane. Proceed as in (ii) and substitute L into the equation of this plane

$$-4(-1-4\lambda)+(0+\lambda)+7(3+7\lambda) = -8 \Rightarrow \lambda = -\frac{1}{2}$$

Hence

$$P': \quad \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} -4 \\ 1 \\ 7 \end{pmatrix} = \begin{pmatrix} 1 \\ -1/2 \\ -1/2 \end{pmatrix} \Rightarrow \overrightarrow{PP'} = \begin{pmatrix} 0 \\ -7/2 \\ 1/2 \end{pmatrix}$$

and the distance is $|\overrightarrow{PP'}| = \frac{5}{2}\sqrt{2}$.

Errors were mostly arithmetic in nature.

(iv) What is the angle at which Π_1 and Π_2 meet? [new]

SOLUTION

Could try $\cos\alpha = \dfrac{\underline{n}_1 \cdot \underline{n}_2}{|\underline{n}_1|\,|\underline{n}_2|}$ where $\underline{n}_{1,2}$ are the normal vectors of the two planes. Or just notice that $\underline{n}_1 \cdot \underline{n}_2 = (2,1,1)\cdot(1,-3,1) = 0$ so the planes are perpendicular and the angle is $\pi/2$.

Errors were due to students not following through the working. In such instances, they failed to realize that the normals of the planes should have been used.

2. (a) Show that for any vectors $\underline{\mathbf{u}}, \underline{\mathbf{v}}, \underline{\mathbf{w}} \in \mathbb{R}^3$, [Seen similar]

$$[(\underline{\mathbf{u}}+\underline{\mathbf{v}}) \times (\underline{\mathbf{v}}+\underline{\mathbf{w}})] \cdot (\underline{\mathbf{w}}+\underline{\mathbf{u}}) = 2(\underline{\mathbf{u}} \times \underline{\mathbf{v}}) \cdot \underline{\mathbf{w}} \qquad (*)$$

SOLUTION Begin with

$$(\underline{\mathbf{u}}+\underline{\mathbf{v}}) \times (\underline{\mathbf{v}}+\underline{\mathbf{w}}) = \underline{\mathbf{u}} \times \underline{\mathbf{v}} + \underline{\mathbf{v}} \times \underline{\mathbf{v}} + \underline{\mathbf{u}} \times \underline{\mathbf{w}} + \underline{\mathbf{v}} \times \underline{\mathbf{w}}$$

and as $\underline{\mathbf{v}} \times \underline{\mathbf{v}} = 0$ we can write the LHS of (*) as

$[\underline{\mathbf{u}} \times \underline{\mathbf{v}} + \underline{\mathbf{u}} \times \underline{\mathbf{w}} + \underline{\mathbf{v}} \times \underline{\mathbf{w}}] \cdot (\underline{\mathbf{w}}+\underline{\mathbf{u}}) = (\underline{\mathbf{u}} \times \underline{\mathbf{v}}) \cdot \underline{\mathbf{w}} + (\underline{\mathbf{v}} \times \underline{\mathbf{w}}) \cdot \underline{\mathbf{u}}$ + several zero-terms of the form $(\underline{\mathbf{u}} \times \underline{\mathbf{w}}) \cdot \underline{\mathbf{w}} = 0$.

The cyclical property of the triple scalar product gives that

$$\underline{\mathbf{u}} \cdot (\underline{\mathbf{v}} \times \underline{\mathbf{w}}) = \underline{\mathbf{w}} \cdot (\underline{\mathbf{u}} \times \underline{\mathbf{v}})$$

and as the scalar product is commutative, we can rewrite the LHS as $2\underline{\mathbf{u}} \cdot (\underline{\mathbf{v}} \times \underline{\mathbf{w}})$, equal to the RHS, as required.

Students who got it wrong either did not know how to proceed or they assumed that the vectors were 3-D when they are supposed to use the identities.

(b) Consider the matrices

$$A = \begin{pmatrix} 1 & -1 & 1 \\ 3 & -1 & 2 \\ 1 & -2 & 2 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 3 & -2 & 1 \\ -2 & 3 & 1 \\ 1 & -3 & 2 \end{pmatrix}.$$

Obtain (i) $\det(A)$, (ii) $\det(B)$, (iii) $\det(AB)$, and (iv) $\det(BA)$. [seen similar]

SOLUTION Expanding by the first row both times:

(i) $\det(A) = 1(-2+4)+1(6-2)+1(-6+1) = 1$

(ii) $\det(B) = 3(6+3)+2(-4-1)+1(6-3) = 20$

(iii) $\det(AB)$ =$\det(A)\det(B) = 1(20) = 20$ and so (iv) $\det(BA) = 20$ as well.

Surprisingly, a good number of students did not know how to compute the determinant.

(c) You are given a system of linear equations

$$\begin{array}{rcrcrcl} 3x & + & y & - & z & = & \alpha \\ x & - & 3y & + & 2z & = & 5 \\ \beta x & + & 2y & - & 3z & = & 15 \end{array}$$

where α and β are constants.

(i) Write the system in the form $A\underline{\mathbf{x}} = \underline{\mathbf{b}}$ and find $\det(A)$. Hence find the conditions on α, β required so that the sytem has (I) infinitely many solutions , (II) no solutions , and (III) a unique solution. In case (I) find all solutions. [seen similar]

SOLUTION

In matrix form the system is written as $\begin{pmatrix} 3 & 1 & -1 \\ 1 & -3 & 2 \\ \beta & 2 & -3 \end{pmatrix} \underline{\mathbf{x}} = \begin{pmatrix} \alpha \\ 5 \\ 15 \end{pmatrix}$

The augmented coefficient matrix is

$$(A : \underline{\mathbf{b}}) = \begin{pmatrix} 3 & 1 & -1 & \alpha \\ 1 & -3 & 2 & 5 \\ \beta & 2 & -3 & 15 \end{pmatrix}$$

so $\det(A) = 3(9-4) - (-3-2\beta) - (2+3\beta) = 16 - \beta$

For (III), a unique solution, we require A invertible, so $\det(A) \neq 0 \Rightarrow \beta \neq 16$, for any α.

For (I) and (II) we have singular A, so $\beta = 16$. Now use Gaussian elimination, beginning with exchanging the first two rows

$$(A : \underline{\mathbf{b}}) \overset{R_1 \leftrightarrow R_2}{\sim} \begin{pmatrix} 1 & -3 & 2 & 5 \\ 3 & 1 & -1 & \alpha \\ 16 & 2 & -3 & 15 \end{pmatrix} \overset{\substack{R_2 - 3R_1 \\ R_3 - 16R_1}}{\sim} \begin{pmatrix} 0 & -3 & 2 & 5 \\ 0 & 10 & -7 & \alpha - 15 \\ 0 & 50 & -35 & -65 \end{pmatrix} \overset{R_3 - 5R_2}{\sim} \begin{pmatrix} 0 & -3 & 2 & 5 \\ 0 & 10 & -7 & \alpha - 15 \\ 0 & 0 & 0 & 10 - 5\alpha \end{pmatrix}$$

The last row represents $0x + 0y + 0z = 10 - 5\alpha$ so if $\alpha \neq 2$ and $\beta = 16$ the system is inconsistent: case (II), no solutions,

Similarly, if $\alpha = 2$ and $\beta = 16$ there are infinitely many solutions, case (I).

In that case, R_2 gives $10y - 7z = -13$ and there is one free variable.

Choose $z = \lambda \in \mathbb{R}$, then $y = \frac{1}{10}(-13 + 7\lambda)$ and $x = 5 + 3y - 2z = \frac{1}{10}(11 + \lambda)$

The solution is a line in $\mathbb{R}^3$:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{10}\begin{pmatrix} 11 \\ -13 \\ 0 \end{pmatrix} + \frac{\lambda}{10}\begin{pmatrix} 1 \\ 7 \\ 1 \end{pmatrix}$$

Errors took the form of wrong determinant and failure to consider each of the 3 cases carefully.

(ii) For $\beta = 8$, find the inverse matrix A^{-1}. [seen similar]

SOLUTION

Now have $\det(A) = 16 - \beta = 8$ and $A = \begin{pmatrix} 3 & 1 & -1 \\ 1 & -3 & 2 \\ 8 & 2 & -3 \end{pmatrix}$. We obtain minors:

$$M_{11} = \begin{vmatrix} -3 & 2 \\ 2 & -3 \end{vmatrix} = 9 - 4 = 5\,, \quad M_{12} = \begin{vmatrix} 1 & 2 \\ 8 & -3 \end{vmatrix} = -3 - 16 = -19\,,$$

$M_{13} = \begin{vmatrix} 1 & -3 \\ 8 & 2 \end{vmatrix} = 2 + 24 = 26$, etc. Now form the matrices of minors, cofactors:

$$M_{ij} = \begin{pmatrix} 5 & -19 & 26 \\ -1 & -1 & -2 \\ -1 & 7 & -10 \end{pmatrix} \Rightarrow Cij = (-1)^{i+j} M_{ij} = \begin{pmatrix} 5 & -9 & 26 \\ 1 & -1 & 2 \\ -1 & -7 & -10 \end{pmatrix}$$

Now the adjoint and inverse

$$\text{adj}(A) = C_{ij}^T = \begin{pmatrix} 5 & 1 & -1 \\ -9 & -1 & -7 \\ 26 & 2 & -10 \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{|A|}\,\text{adj}\,(A) = \frac{1}{8}\begin{pmatrix} 5 & 1 & -1 \\ -9 & -1 & -7 \\ 26 & 2 & -10 \end{pmatrix}$$

Many students employed Gaussian elimination and a lot of arithmetical errors could be found.

(d) Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be a 2×2 matrix. Considering the cases where b or c, or both, are zero, or non-zero, find all possible A satisfying [new]

$$A = A^{-1}.$$

SOLUTION Begin with $A = A^{-1} \Rightarrow A^2 = I$, the identity matrix, which simplifies the equations involved.

$$A^2 = \begin{pmatrix} a^2+bc & b(a+d) \\ c(a+d) & d^2+bc \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Following the hint, begin with

(i) let $b = c = 0$, then $A^2 = \begin{pmatrix} a^2 & 0 \\ 0 & d^2 \end{pmatrix} \Rightarrow a = \pm 1, d = \pm 1 \Rightarrow A = \begin{pmatrix} \pm 1 & 0 \\ 0 & \pm 1 \end{pmatrix}$

(ii) Now let $b = 0$ and $c \neq 0$. Then $A^2 = \begin{pmatrix} a^2 & 0 \\ c(a+d) & d^2 \end{pmatrix}$

Again $a = \pm 1, d = \pm 1$ but in addition, $c(a+d) = 0$ and as $c \neq 0$, must have $a = -d$ so

$A = \begin{pmatrix} 1 & 0 \\ c & -1 \end{pmatrix}$ or $\begin{pmatrix} -1 & 0 \\ c & 1 \end{pmatrix}$, for any non-zero c.

(iii) Finally, both $b \neq 0$ and $c \neq 0$. As in (ii), this implies $a = -d$. As now

$a^2 + bc = 1$ and $d^2 + bc = 1$ are equivalent, leaving only $c = \dfrac{1-a^2}{b} \neq 0$ so

$A = \begin{pmatrix} a & b \\ \frac{1-a^2}{b} & -a \end{pmatrix}$ for any $b \neq 0$ and $a \neq \pm 1$.

Many students directly tackled the problem, i.e. trying to find conditions for $A = A^{-1}$. The result was insufficient details presented in the solution.

3. (a) Given the 2×2 matrix $A = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$,

(i) Find the eigenvalues and eigenvectors for A; [4]

SOLUTION

To find the eigenvalues:

$$\det(A - \lambda I) = \begin{vmatrix} 3-\lambda & -1 \\ -1 & 3-\lambda \end{vmatrix} = (3-\lambda)^2 - 1 = \lambda^2 - 6\lambda + 8 = 0 \Rightarrow \lambda = 2, 4\,.$$

To find the eigenvectors, solve $(A - \lambda I)\underline{\mathbf{x}} = \underline{\mathbf{0}}$.

For $\lambda = 2$: $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \underline{\mathbf{x}} = \underline{\mathbf{0}} \Rightarrow x = y$ and the eigenvector is $\underline{\mathbf{x}}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

For $\lambda = 4$: $\begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix} \underline{\mathbf{x}} = \underline{\mathbf{0}} \Rightarrow x = -y$ and the eigenvector is $\underline{\mathbf{x}}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

Most common error observed was the wrong calculation for determinant.

(ii) Hence find an invertible matrix P, and a diagonal matrix D satisfying [3]

$$A = PDP^{-1} \quad \text{and} \quad P^{-1} = P^T\,;$$

SOLUTION

For P we first normalize the eigenvectors: $\hat{\underline{\mathbf{x}}}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, and $\hat{\underline{\mathbf{x}}}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

Hence P is formed by taking the normalized eigenvectors as columns, and D has the eigenvalues in the corresponding order:

$$P = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad \text{and} \quad d = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}.$$

Most common error observed was not aligning the i-th element of diagonal matrix D with the i-th column of matrix P.

(b) Find values of α and β so that the first order ODE

$$(2x+\alpha te^x+\beta\cos t)\,\frac{dx}{dt} \;=\; -e^x+x\sin t+2t$$

is exact. Hence find the solution satisfying $x(\pi)=0$. [seen similar]

SOLUTION

Rewrite first in standard form $P(t,x)+Q(t,x)\dfrac{dx}{dt}=0$:

$$e^x-x\sin t-2t+(2x+\alpha te^x+\beta\cos t)\,\frac{dx}{dt}=0$$

And for an exact equation, we need the partial derivatives to be equal: $P_x=Q_t$:

$$\frac{\partial}{\partial x}(e^x-x\sin t-2t)=e^x-\sin t \quad\text{and}\quad \frac{\partial}{\partial t}(2x+\alpha te^x+\beta\cos t)=\alpha e^x-\beta\sin t\,,$$

so the ODE is exact with $\alpha=\beta=1$. With these values, there exists a function $f(x,t)$ with $P=f_t$ and $Q=f_x$, so that

$$P=\frac{\partial f}{\partial t}\Rightarrow f=\int P\,dt=\int e^x-x\sin t-2t\;dt=te^x+x\cos t-t^2+g(x)$$

and similarly

$$Q=\frac{\partial f}{\partial}\Rightarrow f=\int Q\,dx=\int 2x+te^x+\cos t\;dx=x^2+te^x+x\cos t+h(t)$$

where h and g are arbitrary functions. Comparing we have $h(t)=-t^2$ and $g(x)=x^2$ giving

$f(t,x)=x^2+te^x+x\cos t-t^2$ with the ODE rewrittten as $df=0$ so the solution is given by $f=$constant:

$$x^2+te^x+x\cos t-t^2=C$$

and the initial condition $x(\pi)=0$ gives $0^2+\pi e^0+0\cos\pi-\pi^2=C$ so $C=\pi-\pi^2$.

Careless mistakes in finding P and Q.

(c) Solve the Bernoulli equation

$$\frac{dy}{dx}-\frac{2}{x}y=x^2y^4$$

satisfying $y(1) = 1$, by using the substitution $u = \dfrac{1}{y^3}$. [seen similar]

SOLUTION

Following the hint, $u = y^{-3} \Rightarrow \dfrac{du}{dx} = -3y^{-4}\dfrac{dy}{dx}$ and we multiply the ODE by $-3y^{-4}$ so that

$$-3y^{-4}\frac{dy}{dx} + \frac{6}{x}y^{-3} = -3x^2 \Rightarrow \frac{du}{dx} + \frac{6}{x}u = -3x^2$$

This is now a linear first-order ODE with integrating factor $e^{\int \frac{6}{x}\,dx} = x^6$ so we multiply by x^6 to obtain

$$x^6\frac{du}{dx} + 6x^5u = \frac{d}{dx}(ux^6) = -3x^8 \Rightarrow ux^6 = -3\int x^8\,dx = -\frac{1}{3}x^9 + C \Rightarrow u = -\frac{1}{3}x^3 + \frac{C}{x^6}$$

And substituting back for $u = y^{-3}$ we have

$$y^{-3} = -\frac{1}{3}x^3 + \frac{C}{x^6} \Rightarrow y = \left(-\frac{1}{3}x^3 + \frac{C}{x^6}\right)^{-1/3}$$

Finally, the initial condition $y(1) = 1$ gives $1^{-3} = (-1/3 + C) \Rightarrow C = 4/3$ and the solution is

$$y = \left(-\frac{1}{3}x^3 + \frac{4}{3x^6}\right)^{-1/3}.$$

Students used the substitution but little errors here and there (possibly due to bad handwriting) which led to mistakes

(d) Find the general solution of the second-order inhomogenous ordinary differential equation [seen similar]

$$\frac{d^2x}{dt^2} + 4x = 2\sin(2t).$$

SOLUTION The characteristic equation is $\lambda^2 + 4 = 0 \Rightarrow \lambda = \pm 2i$ so the complementary function is

$$x_c = c_1\cos 2t + c_2\sin 2t.$$

The input $2\sin 2t$ is in x_c so for the particular integral we try

$x_p = t(a\cos 2t + b\sin 2t) \Rightarrow$

$\dot{x}_p = -2at\sin 2t + a\cos 2t + 2bt\cos 2t + b\sin 2t \Rightarrow$

$\ddot{x}_p = -4at\cos 2t - 4a\sin 2t - 4bt\sin 2t + 4b\cos 2t$

and substituting into the ODE we have

$$\ddot{x_p} + 4x_p = -4at\cos 2t - 4a\sin 2t - 4bt\sin 2t + 4b\cos 2t + 4t(a\cos 2t + b\sin 2t) = 4b\cos 2t - 4a\sin 2t = 2\sin 2t$$

so the undeterminded coefficients are $a = -1/2$ and $b = 0$ giving the PI $x_p = -\frac{1}{2}\sin 2t$ and the general solution is

$$x(t) = c_1\cos 2t + c_2\sin 2t - \frac{1}{2}\sin 2t\,.$$

Careless mistakes in finding the complementary solution.

4. (a) Show that the substitution $x = e^t$ transforms the Euler equation

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + 4y = \frac{1}{x}$$

into the second order linear ODE with constant coefficients

$$\frac{d^2y}{dt^2} + A\frac{dy}{dt} + By = f(t)\,,$$

where A, B and $f(t)$ are to be determined. [bookwork]

SOLUTION

Using $x = e^t \Rightarrow \dfrac{dx}{dt} = x, \quad \dfrac{d}{dt}\left(x^{-1}\right) = -x^{-1}$. Then

$$\begin{aligned}
\frac{dy}{dx} &= \frac{dy}{dt}\left(\frac{dx}{dt}\right)^{-1} = \frac{dy}{dt}x^{-1} \Rightarrow \boxed{x\frac{dy}{dx} = \frac{dy}{dt}} \\
\frac{d^2y}{dx^2} &= \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}\left(\frac{dy}{dt}x^{-1}\right) = \frac{d}{dt}\left(\frac{dy}{dt}x^{-1}\right)\left(\frac{dx}{dt}\right)^{-1} \\
&= \left(\frac{d^2y}{dt^2}x^{-1} - \frac{dy}{dt}x^{-1}\right)x^{-1} \Rightarrow \boxed{x^2\frac{d^2y}{dx^2} = \frac{d^2y}{dt^2} - \frac{dy}{dt}}
\end{aligned}$$

Substituting into the ODE we get

$$\frac{d^2y}{dt^2} - \frac{dy}{dt} + \frac{dy}{dt} + 4y = \frac{d^2y}{dt^2} + 4y = e^{-t}\,,$$

so that $A = 0\,, B = 4$ and $f(t) = e^{-t}$.

Hence solve first for $y(t)$ and thus obtain the solution for $y(x)$. [seen similar]

SOLUTION

For $y(t)$ note $y_c = c_1\cos 2t + c_2\sin 2t$, see Q3d. For the PI we try $y_p = ae^{-t} \Rightarrow \ddot{y}_p = ae^{-t}$ and substitute into the ODE to get $ae^{-t} + 4ae^{-t} = e^{-t} \Rightarrow a = \dfrac{1}{5}$ and the solution is

$$y = c_1\cos 2t + c_2\sin 2t + \frac{1}{5}e^{-t} = c_1\cos(2\ln x) + c_2\sin(2\ln x) + \frac{1}{5x}\,.$$

Errors were in two categories:
(1) the complementary solution was $c_1\cos(2t)+ic_2\sin(2t)$. Not sure why they had the extra "i" in the complementary solution.
(2) students made mistakes from not obtaining the alternate form of the solution $c_1\cos(2\ln x)+c_2\sin(2\ln x)+\frac{1}{5x}$: $Ax^{2i}+Bx^{-2i}+\frac{1}{5x}$.

(b) Find the Taylor series approximation of

$$f(x,y)=\ln(1+x^2-y)$$

up to quadratic terms, about the point $(0,0)$. [seen similar]

SOLUTION For the expansion to quadratic terms, require $f(0,0)=0$, and first-order partial derivatives;

$$f_x=\frac{2x}{1+x^2-y}\Rightarrow f_x(0,0)=0\,,\quad f_y=-\frac{1}{1+x^2-y}\Rightarrow f_y(0,0)-1\,.$$

and second-order partial derivatives

$$f_{xx}=\frac{2-2x^2-2y^2}{(1+x^2-y)^2}\Rightarrow f_{xx}(0,0)=2\,,\quad f_{xy}=\frac{2x}{(1+x^2-y)^2}\Rightarrow f_{xy}(0,0)=0\,,$$

and $f_{yy}=-\dfrac{1}{(1+x^2-y)^2}\Rightarrow f_{yy}(0,0)=-1\,.$

Hence the Taylor series is

$$f(x,y)=f(0,0)+f_x(0,0)x+f_y(0,0)y+\tfrac{1}{2}f_{xx}(0,0)x^2+f_{xy}(0,0)xy+\tfrac{1}{2}f_{yy}(0,0)y^2+\text{h.o.t.}$$

$$=-y+x^2-\tfrac{1}{2}y^2 \text{ up to quadratic terms.}$$

If we choose $x=0.1$ and $y=-0.1$ to estimate $\ln(1.11)$, we expect this to be correct to how many decimal places? [new]

SOLUTION

To estimate $\ln(1.11)$ with $x=0.1$ and $y=-0.1$ we note

$$\ln(1+x^2-y)=\ln(1+0.1^2-(-0.1))=\ln(1.11)$$

and we can use the expansion

$$\ln(1.11) = -(-0.1) + 0.1^2 - (-0.1)^2 = 0.1$$

with an error given by the discarded terms of order x^3, x^2y, xy^2, y^3 in each case, size $0.1^3 = 0.001$, so we would expect the estimate to be accurate to two decimal places.

Errors were either arithmetic or fundamental in nature. By the latter, it means that students didn't tackle it properly from the correct principles.

(c) Obtain all stationary points of the function

$$f(x,y) = 1 - \cos x + \frac{1}{2}y^2,$$

and classify them using the Hessian matrix. [seen similar/new]

SOLUTION

Begin by setting the gradient equal to zero: $\nabla f = (f_x, f_y) = (\sin x, y) = (0,0)$.

Hence two equations must be satisfied: $y = 0$ and $\sin x = 0$, giving infinitely many stationary points $(n\pi, 0)$ for any integer n.

To classify with the Hessian, require the second-order derivatives $f_{xx} = \cos x$, $f_{xy} = 0$, and $f_{yy} = 1$.

The Hessian matrix is $H = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = \begin{pmatrix} \cos x & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \det(H) = \cos x$. Evaluating at the stationary points we have $\det(H) = \cos(n\pi) = \pm 1$.

This is positive for even n, so these are extrema. Given that $f_{yy} = 1 > 0$ always, the points $(2n\pi, 0)$ are minima.

The Hessian is negative for odd n, so the points $([2n+1]\pi, 0)$ are saddle points.

Same comment as for (b).

(d) Let $z = f\left(\frac{x}{y}\right)$. Show that $x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = 0$. [seen similar/new]

SOLUTION Let $u = x/y$, and use the diagram $f - u - \left\{ \begin{array}{l} x \\ y \end{array} \right.$ to apply the chain rule:

$$\frac{\partial f}{\partial x} = \frac{df}{du}\frac{\partial u}{\partial x} = f'(u)\frac{1}{y} \quad \text{and} \quad \frac{\partial f}{\partial y} = \frac{df}{du}\frac{\partial u}{\partial y} = -f'(u)\frac{x}{y^2}\ .$$

Hence

$$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = xf'(u)\frac{1}{y} - yf'(u)\frac{x}{y^2} = 0\,,$$

as required.

Most of the errors involved incorrectly implementing the chain rule.